

GENOMIND-INDIA: A Supercharged AI Platform for Rare Genetic Disease Diagnosis, Pharmacogenomics & Reproductive Risk — Built for India

The Single Biggest Unsolved Problem in Indian Healthcare

India has **3 converging genomic crises** that no single platform addresses:

- **80 million rare disease patients** with 7-year average diagnosis delay
- **Pharmacogenomic blindness** — doctors prescribe drugs without knowing if a patient's genes will cause adverse reactions (1.2 lakh drug-induced deaths/year in India)
- **Reproductive genetic risk** — 1 in 25 Indians is a carrier of a serious genetic disorder, yet pre-marital/prenatal screening is nearly zero in rural areas

No platform in India — or the world — solves all three together with Indian language support, Indian population genomics, and AI explainability. GENOMIND-INDIA is that platform.

Hybrid Deployment Model

Doctors & Hospitals → Web Platform (EMR Integration) Patients & Families → Mobile App (Hindi/Regional Language) ASHA Workers → Lightweight Mobile App (Offline capable) Labs & Diagnostics → REST API integration Researchers → Data dashboard + federated insights

All five user types. One unified backend. This is the hybrid.

THE 11 MODULES

MODULE 1 — Multilingual Clinical NLP Engine

"The Ears of the System"

Problem it solves

Indian doctors write clinical notes in English, Hindi, Hinglish, Tamil-English mix, or pure regional scripts. Zero existing medical NLP tool handles this. Patient symptoms get lost in translation.

What it does

- Ingests free-text clinical notes in 12 Indian languages + code-mixed text
- Performs **Named Entity Recognition (NER)** to extract symptoms, duration, severity, family history, lab values
- Understands Indian-specific symptom descriptions ("haath pair mein sujan", "aankhon ka peela hona")
- Normalizes everything into structured clinical data

ML Technique

- Fine-tuned **MuRIL + BioBERT hybrid** model
- Custom Indian Medical NER dataset built from AIIMS case reports, PubMed Indian studies
- **Code-mixing handler** using language identification at token level (not sentence level — novel)

Why it's novel

Token-level language identification inside medical NER for Indian code-mixed clinical text has never been done. This alone is a publishable paper + patent claim.

MODULE 2 — HPO Phenotype Mapper with Indian Synonym Engine

"The Translator Between Doctors and Databases"

Problem it solves

Global rare disease databases use Human Phenotype Ontology (HPO) terms — standardized English medical codes. Indian clinical descriptions don't map cleanly to these. A doctor writing "progressive proximal muscle weakness with waddling gait" needs to auto-map to HPO:0003560. Currently this is done manually by specialists — there are fewer than 500 in India.

What it does

- Takes extracted clinical entities from Module 1
- Maps them to **HPO ontology terms** automatically
- Handles Indian regional symptom synonyms, Ayurvedic terminology overlap, and colloquial descriptions

- Builds a continuously growing **Indian Medical Synonym Dictionary** — crowdsourced from confirmed cases

ML Technique

- **Semantic similarity using sentence transformers** (fine-tuned on HPO + Indian clinical text pairs)
- Contrastive learning to bring Indian symptom descriptions close to their HPO embeddings in vector space
- Active learning loop — low-confidence mappings flagged for expert review, corrections fed back

Why it's novel

No HPO mapper exists for Indian languages or Indian clinical terminology. The Indian synonym ontology itself is a patentable knowledge artifact.

MODULE 3 — India-Specific Rare Disease Knowledge Graph

"The Brain of the System"

Problem it solves

Global databases like Orphanet and OMIM are built on Western population data. Indian populations have unique variant frequencies, founder mutations in specific communities (e.g., Thalassemia in Sindhis, G6PD in South Indians, Wilson's disease in Andhra Pradesh), and high consanguinity rates in certain states. No knowledge graph encodes this Indian context.

What it does

Builds a massive graph with these interconnected node types:

- Disease nodes (7000+ rare diseases)
- Gene nodes with Indian allele frequencies
- HPO phenotype nodes
- Drug/treatment nodes
- Indian community/ethnicity nodes (Sindhi, Banjara, tribal groups)
- Consanguinity coefficient nodes by state/region
- Lab test nodes with Indian NABL lab availability
- Doctor/specialist nodes by geography

ML Technique

- **Graph construction:** Neo4j + custom ETL pipelines from Orphanet, OMIM, IndiGen, ClinVar, Genome Asia 100K

- **Graph Neural Network (GNN):** PyTorch Geometric — GraphSAGE architecture
- **Link prediction** to discover unknown disease-gene-phenotype connections from Indian data
- **Community detection** to find geographic disease clustering patterns

Why it's novel

The inclusion of Indian ethnicity nodes, consanguinity coefficients, regional founder mutation frequencies, and Indian lab availability as first-class graph entities makes this KG entirely novel and patentable as a knowledge artifact + the GNN trained on it.

MODULE 4 — GNN-Powered Differential Diagnosis Engine

"The Doctor's AI Second Opinion"

Problem it solves

Given a patient's phenotype profile, ranking which rare genetic disease they most likely have — out of 7000+ possibilities — is computationally and cognitively impossible for a general physician. This is why diagnosis takes 7 years. Specialists exist only in metros.

What it does

- Takes HPO terms from Module 2 + patient metadata (age, sex, state, ethnicity, family history)
- Traverses the Knowledge Graph from Module 3 using GNN
- Returns **top-10 ranked differential diagnoses** with:
 - Probability score
 - Which symptoms drove each diagnosis
 - What confirmatory tests to order
 - Expected inheritance pattern
 - Nearest specialist and genetic testing lab in India

ML Technique

- **Heterogeneous GNN** (HAN — Heterogeneous Attention Network) on the multi-node-type KG
- **Phenotype embedding** using graph attention to weight rare/specific symptoms higher than common ones
- **Bayesian uncertainty quantification** — system knows when it doesn't know and says so
- Benchmark against: Phenomizer, MOON, Exomiser (all Western tools)

Why it's novel

No GNN-based differential diagnosis tool exists that uses an India-specific knowledge graph with population-aware node weights. The Bayesian uncertainty layer on top of GNN for rare disease diagnosis is novel.

MODULE 5 — Pharmacogenomics Risk Alert Engine

"Stop Dangerous Drug Prescriptions Before They Happen"

Problem it solves

India has 1.2 lakh adverse drug reaction deaths per year. A significant portion are **pharmacogenomically preventable** — meaning the patient's genetic makeup made that drug dangerous for them specifically. Doctors have no way to know this without genetic testing, which nobody does. Certain Indian populations have very high frequencies of dangerous drug-metabolizing gene variants (CYP2C19, CYP2D6, G6PD) that are underrepresented in Western pharmacogenomic databases.

What it does

- Takes patient's available genetic data (if uploaded) OR infers population-level risk from ethnicity/state
- Cross-references with prescribed medications in their clinical notes (extracted by Module 1)
- Flags **HIGH RISK drug-gene interactions** with severity scores
- Suggests safer alternative drugs
- Covers: chemotherapy, psychiatric drugs, cardiac drugs, antiepileptics, antimalarials (huge in India)

ML Technique

- **Indian-population-calibrated pharmacogenomic risk model** trained on PharmGKB + IndiGen variant frequencies
- **Drug-gene interaction graph** with Indian allele frequency weighted edges
- **Population stratification model** — if no genetic data available, uses state + ethnicity to estimate variant probability (novel probabilistic inference approach)
- NLP from Module 1 extracts drug names from clinical notes automatically

Why it's novel

The population-stratified pharmacogenomic risk inference (estimating drug risk without genetic test, purely from demographic + geographic data) using Indian allele frequencies is completely novel and highly patentable.

MODULE 6 — Reproductive & Prenatal Genetic Risk Counselor

"Preventing Genetic Disease in the Next Generation"

Problem it solves

1 in 25 Indians is a carrier of Thalassemia, Sickle Cell, or other autosomal recessive disorders. When two carriers marry, there's a 25% chance of an affected child. Pre-marital genetic counseling is virtually nonexistent outside metros. India has the world's highest number of Thalassemia major children born annually (~10,000/year). This is almost entirely preventable.

What it does

- Takes family history + available lab reports of a couple
- Calculates **carrier probability** for the top 50 autosomal recessive diseases prevalent in India
- Simulates **Punnett square risk scenarios** for their children
- If pregnancy is ongoing — integrates with available prenatal test reports (Triple test, NT scan reports in text/PDF)
- Generates a **risk counseling report in regional language** — explained in plain language, not medical jargon
- Provides nearby genetic counseling centers and government scheme information (Sickle Cell Mission, etc.)

ML Technique

- **Bayesian carrier probability model** using population prevalence priors + family history evidence
- **NLP report parser** — reads uploaded PDF lab reports and extracts relevant values (Module 1 extended)
- **Risk visualization generator** — auto-generates family pedigree diagrams and probability charts
- Multilingual report generation using fine-tuned **IndicBART** (sequence-to-sequence for Indian languages)

Why it's novel

Automated carrier probability estimation using Indian population priors + multilingual plain-language counseling report generation from free-text family history is novel. No such tool exists in any Indian language.

MODULE 7 — Explainable AI (XAI) Diagnosis Justification Layer

"Building Doctor Trust in AI"

Problem it solves

Indian doctors will never trust a black-box AI diagnosis. Without explainability, this system will never be adopted. This is the #1 reason medical AI fails in clinical settings globally — and in India it's even more critical because liability and trust are paramount.

What it does

- For every diagnosis suggestion, generates a **natural language explanation** of why
- Shows which symptoms contributed most to the diagnosis (symptom attribution scores)
- Highlights what is **MISSING** that would make the diagnosis more/less certain
- Shows similar confirmed Indian patient cases (anonymized) that match the current profile
- Generates a **visual phenotype similarity map** — where does this patient sit in the disease space

ML Technique

- **Graph attention visualization** — which nodes/edges in the KG most activated for this diagnosis
- **SHAP values** for feature attribution on structured inputs
- **BERTViz attention maps** for NLP explanation of which clinical text phrases drove which HPO terms
- **Case-based reasoning retrieval** — vector similarity search over anonymized confirmed Indian cases using FAISS
- Explanation text generated by fine-tuned LLM with structured input → narrative output training

Why it's novel

Multi-layer explainability combining graph attention + SHAP + case-based reasoning + narrative generation in Indian languages for rare disease diagnosis is entirely novel. Explainable medical AI in Indian languages has zero prior art.

MODULE 8 — Federated Learning & Privacy-Preserving Hospital Network

"Making the System Smarter Without Exposing Patient Data"

Problem it solves

To improve, this system needs data from thousands of patients across India. But patient genomic data is the most sensitive data that exists. Hospitals won't share it. Patients won't consent to central storage. Without a privacy solution, the model stays weak.

What it does

- Hospitals run the model **locally on their own servers**
- Only **model gradients** (not patient data) are shared with the central server
- Central server aggregates gradients using **Federated Averaging** to improve the global model
- Each hospital's data never leaves their premises
- Builds India's first **distributed rare disease epidemiology map** — which diseases cluster where, without any individual patient data being centralized

- Differential privacy noise added to gradients before sharing

ML Technique

- **Federated Learning:** Flower (flwr) framework
- **Differential Privacy:** Opacus (PyTorch) — mathematically provable privacy guarantees
- **Secure Aggregation:** Cryptographic protocols so even the central server can't reconstruct individual hospital data
- **Non-IID data handling** — hospitals have very different patient populations, model must handle this distribution shift

Why it's novel

Federated rare disease learning across Indian hospitals with non-IID population handling (tribal hospitals vs metro hospitals have completely different disease distributions) with differential privacy is novel. Building a national rare disease map as a federated learning byproduct is a novel application.

MODULE 9 — ASHA Worker & Rural Health Interface

"Taking Genomic Intelligence to the Last Mile"

Problem it solves

90% of India's rare disease patients are in rural areas. ASHA workers are the first point of healthcare contact for 700 million rural Indians. They are not doctors. They cannot read complex reports. They have basic Android phones and often no internet. Yet they could be the most powerful early detection network in the world if given the right tool.

What it does

- **Simplified mobile app** (works on 2G/offline)
- ASHA worker answers a **voice-guided symptom questionnaire** in their local language (spoken, not typed)
- System generates a **triage score**: Green (monitor), Yellow (refer to PHC), Red (urgent genetic specialist referral)
- Generates a **pre-filled referral letter** in the doctor's language with the patient's symptom summary
- Works fully offline — syncs when connectivity available
- Flags **community-level alerts** — if 3+ children in a village show similar symptoms, system alerts district health officer

ML Technique

- **Automatic Speech Recognition (ASR):** Fine-tuned Whisper model on Indian rural accents and dialects

- **Lightweight on-device NLP:** Quantized/distilled version of Module 1's NER model (runs on 2GB RAM Android)
- **Community anomaly detection:** Federated alerts using privacy-preserving aggregation from Module 8
- **Offline-first architecture:** Local SQLite + sync protocol

Why it's novel

Voice-driven genomic triage in Indian rural dialects with community-level genetic disease clustering alerts delivered through ASHA worker interface is completely novel. This bridges genomic medicine and community health workers in a way never attempted anywhere in the world.

MODULE 10 — Continuous Learning & Active Annotation Pipeline

"The System That Gets Smarter Every Day"

Problem it solves

Medical AI models go stale. New diseases are discovered. New Indian variants are published. Confirmed diagnoses from hospitals are a goldmine of training data that currently goes to waste. Without a systematic learning pipeline, the system becomes outdated within a year.

What it does

- Doctors can **confirm or reject** AI diagnosis suggestions — this feedback is captured
- Confirmed cases are **automatically preprocessed** into training examples
- **Uncertainty sampling** — cases where the model was uncertain are prioritized for expert annotation
- New published papers on Indian rare diseases are **automatically scraped, parsed, and integrated** into the KG
- Monthly model retraining cycles with version control
- **Concept drift detection** — if disease patterns shift (e.g., post-pandemic), system detects and flags it

ML Technique

- **Active Learning:** Uncertainty sampling + query-by-committee strategies
- **Automated Knowledge Graph Update:** NLP pipeline that reads new PubMed papers → extracts disease-gene-phenotype triples → proposes KG additions for expert review
- **Online learning layer** for lightweight continuous updates between major retraining cycles
- **MLflow / DVC** for model versioning and experiment tracking

Why it's novel

Automated KG updating from Indian genomic literature combined with active learning from clinician feedback in a federated setting is novel. The concept drift detection specifically for rare disease pattern shifts in Indian populations is novel.

MODULE 11 — National Rare Disease Intelligence Dashboard

"From Individual Patients to National Policy"

Problem it solves

India's government has a **National Policy for Rare Diseases (2021)** but zero real-time data on where diseases are, how many patients exist, which regions need specialists, and which diseases are being missed. Policy is made blind. Research funding is allocated without data. This is why rare diseases remain neglected.

What it does

- **Anonymized, aggregated, privacy-safe** national dashboard
- Real-time maps showing disease prevalence by district, state, ethnicity
- Specialist gap analysis — where are patients but no doctors
- Diagnostic delay tracking — average time to diagnosis improving or worsening
- Drug availability vs pharmacogenomic need mapping
- **Research opportunity identifier** — which Indian-prevalent diseases have the least research attention (research gap score)
- Exportable reports for Ministry of Health, ICMR, DBT policy briefs

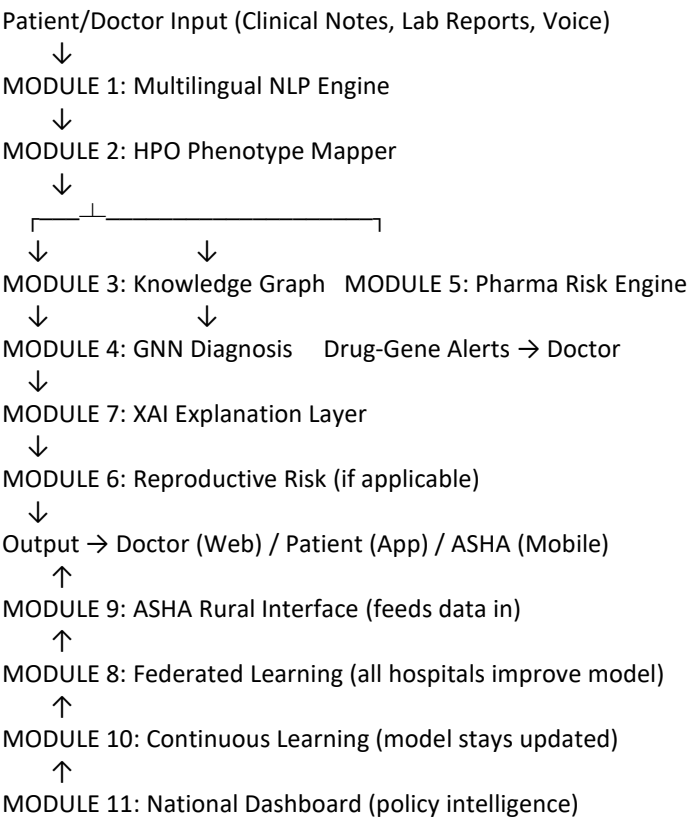
ML Technique

- **Differential privacy aggregation** from Module 8's federated network
- **Geospatial ML** — disease clustering analysis using spatial autocorrelation (Moran's I + ML extensions)
- **Forecasting models** — predict which districts will see rising rare disease burden based on demographic trends
- **NLP policy brief generator** — automatically drafts government-style reports from dashboard data

Why it's novel

A real-time, federated, privacy-preserving national rare disease intelligence system that feeds directly into government policy — with automated policy brief generation — has never been built anywhere in the world, let alone India.

HOW ALL 11 MODULES CONNECT



Patent Claims Summary (14 Novel Claims)

1. Token-level multilingual code-mixed medical NER for Indian languages
 2. Indian medical synonym ontology for HPO mapping
 3. India-specific rare disease KG with consanguinity + ethnicity nodes
 4. Population-stratified pharmacogenomic risk inference without genetic testing
 5. GNN differential diagnosis using Indian-population-weighted heterogeneous graph
 6. Bayesian carrier probability estimation using Indian population priors
 7. Multi-layer XAI combining graph attention + SHAP + case-based reasoning in Indian languages
 8. Federated rare disease learning with non-IID Indian hospital population handling
 9. Voice-driven genomic triage for rural Indian dialects on offline mobile
 10. Community-level genetic disease clustering alerts through ASHA network
 11. Automated KG updating from Indian genomic literature using NLP triple extraction
 12. Concept drift detection for rare disease pattern shifts in Indian populations
 13. Privacy-preserving national rare disease epidemiology map via federated aggregation
 14. Automated government policy brief generation from federated health intelligence
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Tech Stack

Layer	Technology
NLP	HuggingFace, MuRIL, BioMistral, IndicBART, Whisper
Graph	Neo4j, PyTorch Geometric, PyKEEN, NetworkX
ML/DL	PyTorch, Scikit-learn, XGBoost
Explainability	SHAP, BERTViz, FAISS
Federated Learning	Flower (flwr), Opacus
Backend	FastAPI, PostgreSQL, Redis
Mobile	Flutter (offline-first), SQLite
Dashboard	React, D3.js, Plotly
MLOps	MLflow, DVC, Docker, Kubernetes